ITA06020 MACHINE LEARNING

Experiment 11 – Credit Score Classification

Aim:

To implement a Credit Score Classification model using a suitable Machine Learning algorithm in Python to classify customers based on their credit information.

Dataset: German Credit Dataset (real-world benchmark dataset available through OpenML).

Algorithm: Logistic Regression.

Procedure

1. Import the required Python libraries.
2. Load the German Credit dataset.
3. Separate input features and target values.
4. Identify numerical and categorical features.
5. Split the dataset into training and testing data.
6. Encode categorical values and normalize numerical features.
7. Create and train the Logistic Regression model.
8. Predict the credit classification and calculate accuracy.

Python Program

```
import pandas as pd

from sklearn.datasets import fetch_openml

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import StandardScaler, OneHotEncoder

from sklearn.compose import ColumnTransformer

from sklearn.pipeline import Pipeline

from sklearn.linear_model import LogisticRegression

from sklearn.metrics import accuracy_score

data = fetch_openml(name='credit-g', version=1, as_frame=True)

X = data.data

y = data.target

num = X.select_dtypes(include=['number']).columns

cat = X.select_dtypes(exclude=['number']).columns

preprocessor = ColumnTransformer([
```

```
    ('num', StandardScaler(), num),
    ('cat', OneHotEncoder(handle_unknown='ignore'), cat)
])
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
model = Pipeline([
    ('preprocess', preprocessor),
    ('classifier', LogisticRegression(max_iter=1000))
])
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("Credit Score Classification")
print("Training Samples:", len(X_train))
print("Testing Samples:", len(X_test))
print("Predicted Credit:", y_pred[0])
print("Actual Credit:", y_test.iloc[0])
print("Accuracy:", round(accuracy_score(y_test, y_pred)*100, 2), "%")
```

Output:



Result

Thus, the Credit Score Classification model was successfully implemented using Logistic Regression and the German Credit dataset to classify customers into good and bad credit categories.